

3.3.2 Refraction, diffraction and interference

3.3.2.1 Interference

Content	Opportunities for skills development
Path difference. Coherence. Interference and diffraction using a laser as a source of monochromatic light. Young's double-slit experiment: the use of two coherent sources or the use of a single source with double slits to produce an interference pattern. Fringe spacing, $w = \frac{\lambda D}{s}$ Production of interference pattern using white light. Students are expected to show awareness of safety issues associated with using lasers. Students will not be required to describe how a laser works. Students will be expected to describe and explain interference produced with sound and electromagnetic waves. Appreciation of how knowledge and understanding of nature of electromagnetic radiation has changed over time.	AT i Investigation of two-source interference with sound, light and microwave radiation.
Required practical 2: Investigation of interference effects to include the Young's slit experiment and interference by a diffraction grating.	

3.3.2.2 Diffraction

Content	Opportunities for skills development
Appearance of the diffraction pattern from a single slit using monochromatic and white light. Qualitative treatment of the variation of the width of the central diffraction maximum with wavelength and slit width. The graph of intensity against angular separation is not required. Plane transmission diffraction grating at normal incidence. Derivation of $d\sin\theta = n\lambda$ Use of the spectrometer will not be tested. Applications of diffraction gratings.	

3.3.2.3 Refraction at a plane surface

Content	Opportunities for skills development
Refractive index of a substance, $n = \frac{c}{c_s}$ Students should recall that the refractive index of air is approximately 1. Snell's law of refraction for a boundary $n_1\sin\theta_1 = n_2\sin\theta_2$ Total internal reflection $\sin\theta_c = \frac{n_2}{n_1}$ Simple treatment of fibre optics including the function of the cladding. Optical fibres will be limited to step index only. Material and modal dispersion. Students are expected to understand the principles and consequences of pulse broadening and absorption.	MS 0.6, 4.1